

Supplementary Information

Pre-cooling a superconducting-qubit dilution refrigerator: cascade heat-budget penalty and helium-versus-nitrogen cold-plate coolant selection by CFD

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Support material—case dictionaries, calibration and gate-check scripts, grid-convergence data, and figure-generation code—is available at <https://github.com/Imerson/Pre-cooling-a-superconducting-qubit-dilution-refrigerator>

S1 Enclosure CHT model: regions, boundary conditions, and calibration

Residual-gas conduction bound. At $p < 10^{-4}$ mbar the Knudsen number satisfies $\text{Kn} \gg 1$, so gas heat transport is free-molecular and follows the Kennard form $\dot{Q}_{\text{gas}} = \alpha \Lambda p A \Delta T$, with Λ the free-molecular conduction coefficient for helium and α the accommodation coefficient^{2,3}. Because $\dot{Q}_{\text{gas}}$ scales linearly with pressure, it evaluates to less than 1% of the conductive stage load at the operating pressure and is reported as a budget line rather than resolved by the continuum solver.

Table S1. Mesh regions, material assignments, and roles for the four enclosure CHT cases. The region structure is identical for all cases; temperatures change per case (Table S2).

Region	Type	Material	Role
Warm plate (top)	solid boundary	polished aluminium	warm axial boundary (Dirichlet T)
OVC wall	solid boundary	polished aluminium	warm radial boundary (Dirichlet T)
domain0 (outer gap)	fluid	near-vacuum He	grey-body radiation transport
Radiation shield	solid boundary	polished 316L SS + Cu coating	first-shield boundary (Dirichlet T)
domain1 (inner gap)	fluid	near-vacuum He	conduction/radiation coupling
Cold plate (base)	solid boundary	OFHC copper	cold axial boundary (Dirichlet T)
Coax bundles ($\times 4$)	solid	lumped composite cylinders	conduction path (per-bundle $\kappa_{\text{eff}}(T)$)

Table S2. Per-case boundary temperatures. Grey-body (view-factor) radiation is active in all four cases; residual-gas viscosity is suppressed ($\mu = 10^{-11}$ Pa s at $p < 10^{-4}$ mbar), velocity is noSlip, and coax–fluid interfaces carry conjugate coupling.

Case	Stage span	Configuration	T_{hot}	T_{cold}
1	Stage 1	He	300 K	50 K
2	Stage 1	LN ₂	300 K	77 K
3	Stage 2	He	50 K	4 K
4	Stage 2	LN ₂	77 K	4 K

Table S3. Calibrated per-bundle conduction loads (mW) reproduced by the temperature-dependent harness $\kappa_{\text{eff}}(T)$ polynomial calibration (per-bundle fifth-order fit, error $< 0.5\%$). Component conductivities are integrated from tabulated cryogenic data^{1,2}. Bundles are scaled by line count: L1 drive (25), L2 flux (25), L3 read (10), L4 pump (5); the flux bundle uses the stainless reference cable, with phosphor-bronze differentiation a pending refinement.

Stage span	L1 (25)	L2 (25)	L3 (10)	L4 (5)	Total
300 $\rightarrow$ 50 K	395.8	395.8	158.3	79.2	1029
300 $\rightarrow$ 77 K	370.0	370.0	148.0	74.0	962
50 $\rightarrow$ 4 K	27.0	27.0	10.8	5.4	70.3
77 $\rightarrow$ 4 K	62.3	62.3	24.9	12.5	162.0

Table S4. Stage-1 (50 K) radiative load: `viewFactor` agglomeration sweep against the exact grey-body anchor (7.07 W = 4.24 W outer gap + 2.84 W inner gap). The sweep is non-monotonic; the production value is retained as a conservative upper bound and the analytic value as the best estimate. Stage 2 corroborates the method (analytic 3.15 mW vs production 3.44 mW).

Mesh / method	Radiation [W]	Note
Analytic grey-body anchor	7.07	exact (best estimate)
Coarse	9.46	under-resolved geometry; over-predicts
Production (medium)	8.41	+19%; conservative upper bound
Over-agglomerated	3.58	view-factor closure lost; artefact

Table S5. Cold-plate heat-exchanger case definitions. Both cases share geometry, mesh, duty, and Reynolds number; only the coolant and its solver differ.

Property	He case (50 K)	LN ₂ case (77 K)
Solver	compressible (<code>rhoThermo</code>)	variable-density <code>buoyantBoussinesqSimpleFoam</code>
Pr	0.77	2.36
k [W m ⁻¹ K ⁻¹]	0.0185	0.140
$\beta\Delta T$ (validity)	0.63 (variable-density required)	0.03 (Boussinesq valid)
Duty $\dot{Q}$ / Re	9.4 W / 2300	9.4 W / 2306
Production mesh	2.92×10^5 cells (graded)	2.92×10^5 cells (graded)

Table S6. Dimensionless groups used in the cold-plate assessment. The fully developed square-duct references are $Nu_\infty = 3.61$ and $f Re = 56.9$ ⁴. The bulk temperature is $T_b = \frac{1}{2}(T_{in} + T_{out})$ and $h = \dot{Q}/[A_{heated}(T_w - T_b)]$.

Group	Definition	Role
Reynolds Re	$\rho U D_h / \mu$	flow regime (matched at 2300)
Nusselt Nu	$h D_h / k$	convective enhancement vs Nu_∞
Friction f	$2\Delta p D_h / (\rho U^2 L)$	hydraulic cost; $f Re$ vs 56.9
NTU, ε	$h A / \dot{m} c_p$, $1 - e^{-NTU}$	exchanger effectiveness ⁵
Mouromtseff Mo	$k^{0.6} \rho^{0.8} c_p^{0.4} \mu^{-0.4}$	coolant figure of merit ⁶
Graetz L_t / L	$0.05 Re Pr D_h / L$	thermal entry length
Axial cond. λ_{ax}	$1/Pe$	axial-conduction significance
Entropy gen. $\dot{S}_{gen}$, Be	$\dot{S}_\theta + \dot{S}_\mu$, $\dot{S}_\theta / \dot{S}_{gen}$	second-law analysis ⁷

Table S7. Complete converged cold-plate metrics, He (50 K) versus LN₂ (77 K), matched 9.4 W at $Re = 2300$. Heat-transfer quantities carry $GCI \approx 2.4\%$; f , Δp , $\dot{W}_p$ carry $GCI \approx 17\%$ (Table S8).

Quantity	He (50 K)	LN ₂ (77 K)	LN ₂ /He
Nu	13.89	33.02	2.38
Nu/Nu _∞	3.85	9.15	2.38
f (Darcy)	0.0798	0.1162	1.46
h [W m ⁻² K ⁻¹]	25.7	460.9	17.9
$T_w - T_b$ [K]	24.4	1.36	0.056
$\dot{W}_p$ [μ W]	143	68.5	0.48
NTU	0.236	0.182	0.77
ε	0.211	0.167	0.79
Mo	470	4.51×10^4	96
L_t/L (Graetz)	2.94	9.07	3.09
λ_{ax}	5.7×10^{-4}	1.8×10^{-4}	0.32
$\dot{S}_{gen}$ [W K ⁻¹]	0.0500	0.00123	0.025
Bejan irreversibility (thermal)	1.000	1.000	—
Exergy destruction [W]	15.0	0.37	0.025
Energy-balance closure	5.2%	2.4%	—

Table S8. Grid Convergence Index study⁸ for the representative LN₂ cold-plate case at $Re = 2300$. Meshes 86k/292k/984k cells, refinement ratio $r = 1.5$. Apparent order p : 1.63 (Nu, h), 0.56 (f). Production mesh: medium (292k); medium-mesh GCI: Nu 2.4%, f 22.8%.

Quantity	Coarse	Medium	Fine	Richardson	GCI _{fine}
Nu	33.61	33.02	31.87	30.64	4.8%
h [W m ⁻² K ⁻¹]	469.2	460.9	444.9	427.8	4.8%
f (Darcy)	0.1108	0.1162	0.1206	0.1374	17.5%

S2 Radiation validation: view-factor agglomeration sweep

S3 Cold-plate model: solvers, definitions, and full metrics

References

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